

FRANCIS SCOTT KEY BRIDGE, MD, USA

WMO: 997302

Lat: 39.218N

Lon: 76.527W

Elev: 33

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 16.0	(c) 20.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 34.4	(k) 31.0	(l) 32.2	(m) 30.2	(n) 15.5	(o) 290	(p) 0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.6	(c) 90.3	(d) N/A	(e) 87.9	(f) N/A	(g) 85.9	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 11.1	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 32.2	(b) 26.7	(c) 23.4	(e) 12.8	(f) 94.6	(g) 4.1	(h) 2.6	(i) 9.8	(j) 96.5	(k) 7.4	(l) 98.1	(m) 5.1	(n) 99.6	(o) 2.1	(p) 101.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.5	35.3	36.5	44.2	56.3	65.4	74.3	79.1	77.3	70.6	59.8	48.5	41.0		
	DBStd	16.90	8.98	8.57	8.53	7.83	6.79	5.27	4.16	4.07	5.43	7.47	7.27	7.64		
	HDD50	1513	461	385	221	27	0	0	0	0	0	10	112	297		
	HDD65	4173	921	797	644	280	81	4	0	0	12	196	494	744		
	CDD50	4241	5	8	43	215	477	729	902	846	618	313	68	17		
	CDD65	1426	0	0	1	18	93	283	437	381	180	33	0	0		
	CDH74	10629	0	2	6	128	430	1944	4121	3013	870	115	0	0		
	CDH80	2856	0	0	0	32	71	487	1335	765	142	24	0	0		
(14) Wind	WSAvg	10.5	11.0	11.5	12.1	11.7	10.9	10.7	9.1	8.9	9.1	10.2	10.4	10.2		
(15) Precipitation	PrecAvg	46.1	3.3	2.8	4.3	3.6	4.0	4.4	4.2	4.4	4.4	3.9	3.0	3.7		
	PrecMax	66.1	5.9	6.4	8.8	7.1	8.3	11.5	7.0	11.1	12.1	8.7	6.0	7.2		
	PrecMin	33.3	1.4	0.6	0.3	1.9	1.3	1.6	1.0	1.5	0.1	0.3	0.8	1.6		
	PrecStd	7.1	1.0	1.5	1.8	1.4	1.7	2.4	1.6	2.1	3.1	2.4	1.5	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.7	68.2	73.3	84.4	86.4	91.0	94.9	92.4	87.8	83.6	69.7	67.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	57.1	58.6	66.1	77.5	81.8	88.0	91.4	88.6	83.7	76.8	65.0	60.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	52.0	52.4	61.0	72.0	78.7	85.2	88.8	86.2	81.3	73.5	62.0	56.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	47.8	47.5	56.5	67.9	75.9	82.5	86.5	84.1	78.9	70.7	59.3	51.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	11.6	12.6	13.7	12.5	11.5	11.6	10.5	10.6	11.4	11.2	10.2		
		MCDBR	18.1	19.6	20.2	19.6	14.8	14.3	14.5	12.8	12.6	14.5	14.8	16.6		
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDWR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.321	0.362	0.400	0.446	0.499	0.518	0.513	0.428	0.363	0.326	0.311		
	taud		2.470	2.427	2.375	2.290	2.196	2.090	2.053	2.051	2.260	2.446	2.510	2.511		
	Ebn at Noon		273	286	283	278	266	251	245	242	258	267	266	265		
	Edh at Noon		25	30	35	41	46	51	52	51	39	29	24	23		
(44) All-Sky Solar Radiation	RadAvg		646	917	1217	1561	1748	1939	1919	1698	1384	1025	739	567		
	RadStd		53	105	92	126	210	149	93	96	161	111	62	55		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (79 neighbors)		+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+232	+131

Nomenclature: See separate page